TECHNICAL DEPTH

Research-Grade Foundations

ROUTING

AODV Protocol

On-demand route discovery — no flooding, no wasted bandwidth

▸ *RFC 3561 Standard*

DETECTION

Bayesian Sensor Fusion

P(disaster | accel + gyro + baro) — multi-sensor posterior probability

▸ *Eliminates false positives*

SECURITY

BFT Consensus

SOS confirmed only after k independent nodes corroborate — fake-proof

▸ *Byzantine Fault Tolerant*

ENERGY

TDMA Sleep Schedule

Time-slotted wake/sleep cycles cut battery draw by up to 80%

▸ *Clock-sync via beacons*

RESILIENCE

Delay-Tolerant Networking

Store-and-forward bridges disconnected mesh segments automatically

▸ *DTN / Bundle Protocol*

SCALE

LEACH Clustering

Rotating cluster-heads aggregate traffic — proven 8× network lifetime gain

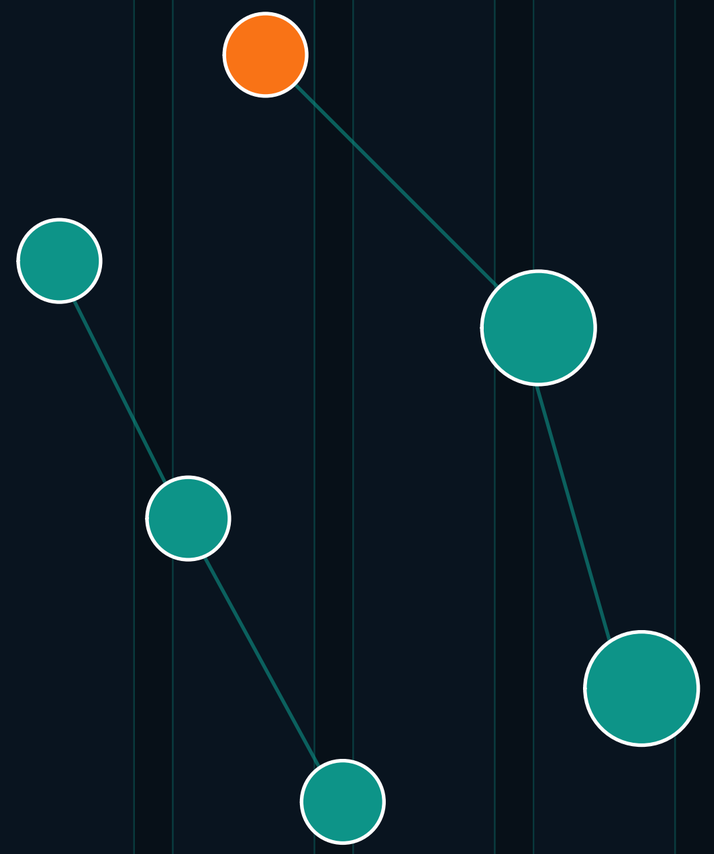
▸ *MIT WSN Research*

EMERGENCY TECH

RescueMesh

Resilient Communication When Infrastructure Fails

Disaster-proof mesh networking • Sensor-triggered SOS • Multi-layer communication



THE PROBLEM

When Disaster Strikes, Communication Fails

72hrs

Average time before cellular networks are restored after major disasters

40%

Rescue missions delayed due to lack of survivor location data

$280B

Annual economic losses from preventable disaster casualties globally

Existing solutions — satellite phones, radio systems — are expensive, specialized, or inaccessible to ordinary people caught in disasters.

THE SOLUTION

RescueMesh: Your Phone Becomes a Lifeline

1. Internet

Primary Mode

↓ AUTOMATIC FALLBACK

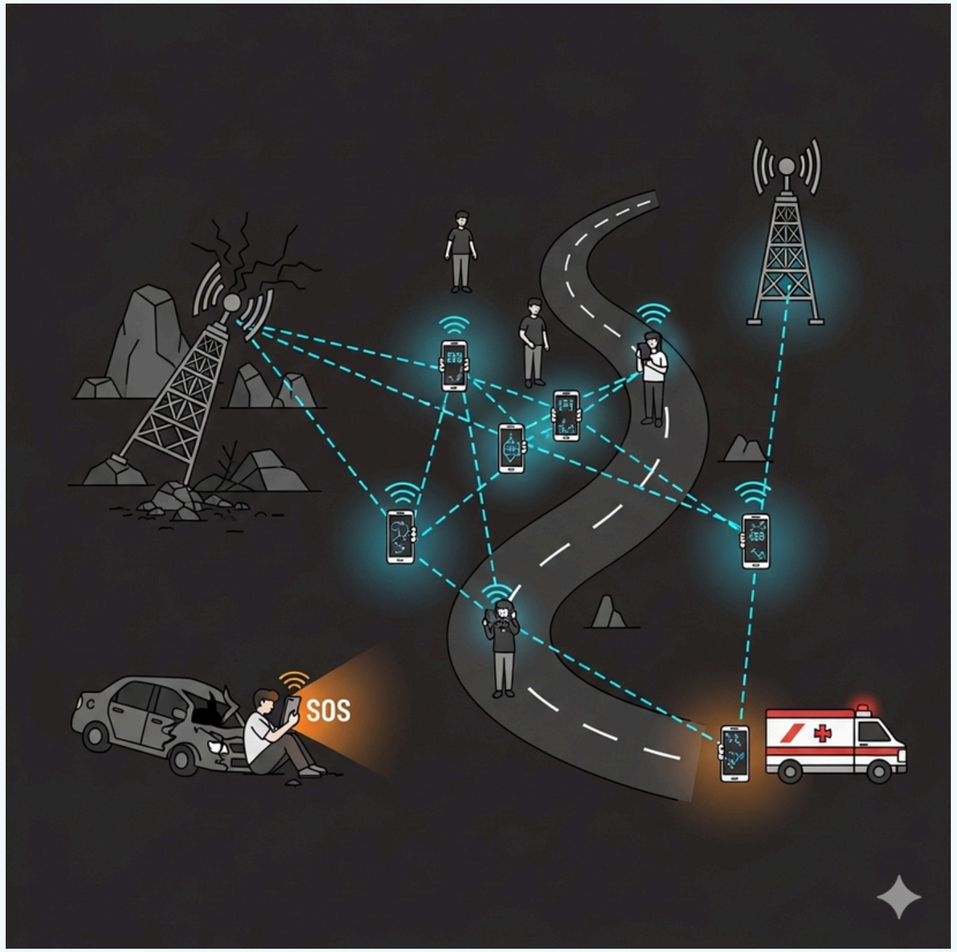
2. Satellite

Fallback Mode

↓ AUTOMATIC FALLBACK

3. Mesh Network

Last Resort



No internet required. No special hardware. Just smartphones.

IMPLEMENTATION PLAN

Full Building Procedure

Vibe-coded. Fully functional. Demo-ready.

Core Mesh Engine

React Native + react-native-ble-plx

- BLE peer discovery & neighbor table
- Beacon packet broadcast (JSON over BLE)
- Basic AODV route table in-memory
- Node ID + GPS + battery in payload

SOS & Detection

expo-sensors + TensorFlow Lite

- Accelerometer threshold + Bayesian layer
- Seismic P-wave pattern classifier
- SOS packet flood with TTL=7
- BFT: require 3 nodes to confirm SOS

Dashboard & Map

Next.js + Leaflet.js + Socket.io

- Real-time survivor location map
- SOS alerts with node heat map
- Mesh topology visualizer
- Gateway node WebSocket bridge

Polish & Demo

Figma + Llama 3.2 1B (on-device)

- Offline AI first-aid chatbot
- TDMA sleep schedule toggle
- Demo simulation video (60 sec)
- Pitch dry-run + Q&A prep

Tech stack: React Native • Next.js • Node.js • Leaflet.js • BLE / Wi-Fi Direct • Socket.io

THE SOLUTION

RescueMesh: Your Phone Becomes a Lifeline

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Last Resort

How It Activates

- Device sensors detect disaster event
- AI classifies emergency type
- Best available network selected
- SOS packet broadcast instantly
- Mesh relays signal hop-by-hop
- Rescuers receive survivor location

No internet required. No special hardware. Just smartphones.

TECHNICAL ARCHITECTURE

Multi-Layer System Design

SENSOR LAYER	Accelerometer • Gyroscope • Barometer • Health Sensors
DETECTION	Crash Detection • Fall Detection • Seismic Analysis • Anomaly Classification
COMM LAYER	Internet API • Satellite Uplink • BLE Mesh
ROUTING LAYER	Gossip Protocol • TTL Packets • Neighbor Table • Flood Routing
AUTHORITY LAYER	Emergency Dashboard • Rescue Coordination

IMPLEMENTATION PLAN

Full Build in 10 Hours

Vibe-coded. Fully functional. Demo-ready.

Hr 0–2

Core Mesh Engine

React Native + react-native-ble-plx

- BLE peer discovery & neighbor table
- Beacon packet broadcast (JSON over BLE)
- Basic AODV route table in-memory
- Node ID + GPS + battery in payload

Hr 2–5

SOS & Detection

expo-sensors + TensorFlow Lite

- Accelerometer threshold + Bayesian layer
- Seismic P-wave pattern classifier
- SOS packet flood with TTL=10
- BFT: require 3 nodes to confirm SOS

Hr 5–7

Dashboard & Map

Next.js + Leaflet.js + Socket.io

- Real-time survivor location map
- SOS alerts with node heat map
- Mesh topology visualizer
- Gateway node WebSocket bridge

Hr 7–10

Polish & Demo

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MESH PROTOCOL

5-Second Beacon Heartbeat System

Beacon Packet Structure

```
node_id:        "A23F"
timestamp:      "1741882123"
gps:            "28.61, 77.20"
battery:        68
signal_dbm:     -62
status:         "SOS" / "normal"
```

Adaptive Ping Intervals

Normal Mode:	Every 30 sec
Emergency Mode:	Every 5 sec
Low Battery:	Every 60 sec

Self-Healing Network

Route auto-recalculates via next-best path

SOS floods with TTL = 7 hops

Node keys prevent fake SOS attacks

BUSINESS MODEL

Who Pays, What They Get

Governments & Disaster Agencies

Improve emergency response, reduce casualties

Multi-year contracts

Software as a Service (SaaS)

Improvement using Softwares and hardwares in phones

Deployment contracts
Partnership pricing

Infrastructure Operators

Emergency comms for airports, stadiums, mines

Enterprise subscription

"RescueMesh sells resilient emergency communication infrastructure to governments and disaster response organizations."

MARKET OPPORTUNITY

$100B+ Global Addressable Market

Government Contracts

Primary Revenue

National & city disaster management

SaaS Platform

Emergency dashboards, heat maps, analytics

Hardware Gateways

Hardware + SW Bundle

Satellite nodes, drone relays, routers

Premium AI Features

Per-Seat Licensing

Predictive evacuation, command tools

Disaster Technology • Emergency Communication • Public Safety Infrastructure • IoT Sensors • AI Platforms

COMPETITIVE ADVANTAGE

Why RescueMesh Wins

Solution	Cost	Requires Special HW	Works Offline	Smartphone Native
Satellite Comms	💰 Very High	✅ Yes	✅ Yes	❌ No
Radio Systems	💰 High	✅ Yes	✅ Yes	❌ No
Internet Apps	💚 Low	❌ No	❌ No	✅ Yes
RescueMesh ⭐	**💚 Low**	**❌ No**	**✅ Yes**	**✅ Yes**

🔋 Battery-aware adaptive beaconing

🔐 Cryptographically signed SOS packets

🤖 On-device AI for disaster detection

🌐 Works on any modern smartphone

EVALUATION CRITERIA

How RescueMesh Scores

💡 Innovation

- First smartphone-native offline mesh for disasters
- Sensor-triggered auto-activation — no human action needed
- Adaptive beaconing protocol (5s emergency / 30s normal)
- Offline AI assistant for first-aid guidance

⚙️ Feasibility

- Built on existing BLE & Wi-Fi Direct — no new hardware
- Proven mesh protocols (gossip, flood routing, TTL)
- Deployable as a lightweight Android/iOS app today
- Pilot-ready for city agencies within 6 months

🎯 Clarity

- Single clear mission: comms when infrastructure fails
- Layered architecture easy to explain and demonstrate
- Defined customer segments with clear revenue model
- Simple user experience — app activates automatically

📈 Scalability

- Each new phone strengthens the network automatically
- Phase roadmap: city → nation → global deployments
- SaaS + contracts model scales without linear cost
- Drone relay + satellite uplink for unlimited range

GROWTH ROADMAP

Phased Path to Market Leadership

Phase 1

App-Based Mesh

- BLE mesh
- Sensor-triggered SOS
- Pilot with city agencies
- App Store launch

Phase 2

Satellite + Drone

- Satellite uplink integration
- Drone relay stations
- NGO partnerships
- Government contracts

Phase 3

AI Disaster Platform

- Predictive AI analytics
- Evacuation optimization
- National deployments
- Global scale

Ready to Save Lives?

RescueMesh

The Technology Already Exists.

Every smartphone today has Bluetooth, Wi-Fi, GPS, and powerful sensors. RescueMesh harnesses this existing hardware to build a life-saving communication network that needs no towers, no internet, and no special equipment.

- ✅ Smartphone-native — no new hardware
- ✅ Auto-activating — no manual action needed
- ✅ Works anywhere on Earth
- ✅ Scales from one city to one nation

RescueMesh by Bitbenders